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Mathematical Foundations of Data Science (Probability)

Lecture 04

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Discrete Random Variable

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Discrete Random Variable)

A random variable $X : \Omega \rightarrow \mathbb{R}$ is said to be **discrete** if there exists a **finite or countably infinite** set $\mathcal{X} \subset \mathbb{R}$ such that

$$\mathbb{P}(\{X \in \mathcal{X}\}) = \mathbb{P}\left(\bigsqcup_{x \in \mathcal{X}} \{X = x\}\right) = \sum_{x \in \mathcal{X}} \mathbb{P}(\{X = x\}) = 1.$$

Remark:

- For any “nice” set $B \subseteq \mathbb{R}$,

$$\mathbb{P}(\{X \in B\}) = \sum_{x \in B} \mathbb{P}(\{X = x\}).$$

Here, “nice” means that $\{X \in B\} \in \mathcal{F}$ so that we can $\mathbb{P}(\{X \in B\})$ makes sense
 In measure theory, such nice sets are called **Borel sets**

Probability Mass Function (PMF)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Probability Mass Function)

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable.

The **probability mass function (PMF)** of X is the function $p_X : \mathbb{R} \rightarrow [0, 1]$ defined as

$$p_X(x) = \mathbb{P}(\{X = x\}), \quad x \in \mathbb{R}.$$

Remark:

- If X is a discrete random variable taking values in a discrete set $\mathcal{X}$, then

$$\sum_{x \in \mathcal{X}} p_X(x) = 1.$$

- If X is a discrete random variable taking values in a discrete set $\mathcal{X}$, then its CDF and PMF satisfy

$$F_X(x) = \mathbb{P}(\{X \leq x\}) = \sum_{\substack{y \in \mathcal{X}: \\ y \leq x}} p_X(y), \quad x \in \mathbb{R}.$$

- For a discrete random variable X , its PMF p_X provides the complete probabilistic description

Examples of PMFs

- Bernoulli(p), $p \in [0, 1]$

$$\mathcal{X} = \{0, 1\},$$

$$p_X(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

- Uniform($\{1, \dots, n\}$), $n \in \mathbb{N}$

$$\mathcal{X} = \{1, \dots, n\},$$

$$p_X(x) = \begin{cases} \frac{1}{n}, & x \in \{1, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

Examples of PMFs

- Geometric(p), $p \in [0, 1]$

$$\mathcal{X} = \mathbb{N}, \quad p_X(x) = \begin{cases} p(1-p)^{x-1}, & x \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases}$$

- Binomial(n, p), $n \in \mathbb{N} \cup \{0\}$, $p \in [0, 1]$

$$\mathcal{X} = \{0, \dots, n\}, \quad p_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x \in \{0, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

Examples of PMFs

- $\text{Poisson}(\lambda)$, $\lambda > 0$

$$\mathcal{X} = \{0, 1, 2, \dots\},$$

$$p_X(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & x \in \{0, 1, 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

- A custom PMF:

$$\mathcal{X} = \{1, 2, \dots\},$$

$$p_X(x) = \begin{cases} \frac{6}{\pi^2} \frac{1}{x^2}, & x \in \{1, 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

Continuous Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Continuous Random Variable)

A random variable $X : \Omega \rightarrow \mathbb{R}$ is said to be **continuous** if there exists a function $f_X : \mathbb{R} \rightarrow [0, \infty)$ such that

$$F_X(x) = \mathbb{P}(\{X \leq x\}) = \int_{-\infty}^x f_X(t) dt, \quad x \in \mathbb{R}.$$

Remarks:

- If $X : \Omega \rightarrow \mathbb{R}$ is a continuous random variable, its CDF F_X is continuous at every point, and therefore

$$\mathbb{P}(\{X = x\}) = 0 \quad \forall x \in \mathbb{R}.$$

- The function f_X in the definition is called the **probability density function (PDF)** of X
- For a continuous random variable X , its PDF f_X provides the full probabilistic description of X

Examples

- $X \sim \text{Uniform}([a, b]), \quad a < b,$

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b], \\ 0, & \text{otherwise.} \end{cases}$$

- $X \sim \text{Exponential}(\lambda), \quad \lambda > 0$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- $X \sim \text{Gaussian}(\mu, \sigma^2), \quad \mu \in \mathbb{R}, \quad \sigma > 0$

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

- $X \sim \text{Normal} = \text{Gaussian}(0, 1)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad x \in \mathbb{R}.$$

PDF $\neq$ Probabilities

- A common mistake students make is to think of PDF values as probability (perhaps because 'P' in PDF stands for probability)
- Recall that CDF and PMF values are defined as probabilities of certain events:

$$F_X(x) = \mathbb{P}(\{X \leq x\}), \quad p_X(x) = \mathbb{P}(\{X = x\}).$$

- However, a PDF value is not a probability, i.e.,

$$f_X(x) \neq \mathbb{P}(\cdot).$$

- The value $f_X(x)$ at a point x represents the **slope** of the CDF F_X at the point x :

$$f_X(x) = \left. \frac{d}{dt} F_X(t) \right|_{t=x}.$$

Note

A probability density function (PDF) does not have the interpretation of a probability. Only integrals of PDF have interpretation of probabilities.

Multiple Random Variables

Joint CDF of Two Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Joint CDF)

Given random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ with respect to $\mathcal{F}$, their **joint CDF**

$$F_{X,Y} : \mathbb{R}^2 \rightarrow [0, 1]$$

is defined as

$$F_{X,Y}(x, y) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}), \quad x, y \in \mathbb{R}.$$

Notations:

- $\{X \leq x\} \cap \{Y \leq y\} = \{X \leq x, Y \leq y\}$
- $\mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}) = \mathbb{P}(\{X \leq x, Y \leq y\})$

Properties of Joint CDF

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ be random variables with respect to $\mathcal{F}$ with joint CDF $F_{X,Y}$

- $\lim_{x,y \rightarrow -\infty} F_{X,Y}(x,y) = 0, \quad \lim_{x,y \rightarrow +\infty} F_{X,Y}(x,y) = 1$

- (Monotonicity)** If $x_1 \leq x_2$ and $y_1 \leq y_2$, then

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2)$$

- $F_{X,Y}$ is **continuous from the right and top**, i.e., for all $x, y \in \mathbb{R}$,

$$\lim_{\varepsilon \downarrow 0, \delta \downarrow 0} F_{X,Y}(x + \varepsilon, y + \delta) = F_{X,Y}(x, y).$$

- $\lim_{y \rightarrow \infty} F_{X,Y}(x, y) = F_X(x)$ for all $x \in \mathbb{R}$
 $\lim_{x \rightarrow \infty} F_{X,Y}(x, y) = F_Y(y)$ for all $y \in \mathbb{R}$

Joint CDF of Multiple Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Joint CDF of More Than 2 Random Variables)

Let $X_1, \dots, X_n$ be random variables with respect to $\mathcal{F}$.

The **joint CDF** of $X_1, \dots, X_n$ is a function $F_{X_1, \dots, X_n} : \mathbb{R}^n \rightarrow [0, 1]$ defined as

$$F_{X_1, \dots, X_n}(\mathbf{x}_1, \dots, \mathbf{x}_n) = \mathbb{P} \left(\bigcap_{i=1}^n \{X_i \leq x_i\} \right), \quad \mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}.$$

Independence of Random Variables

Independence of Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Independence of Random Variables)

1. Two random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ defined with respect to $\mathcal{F}$ are said to be **independent** if

$$F_{X,Y}(x, y) = F_X(x) \cdot F_Y(y) \quad \forall x, y \in \mathbb{R}.$$

2. A collection of random variables $X_1, \dots, X_n$, all defined with respect to $\mathcal{F}$, are said to be independent if

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n), \quad x_1, \dots, x_n \in \mathbb{R}.$$

Can a Random Variable be Independent of Itself?

Can a random variable X be independent of itself?

- For X to be independent of itself, we must have

$$F_{X,X}(x, y) = F_X(x) \cdot F_X(y) \quad \forall x, y \in \mathbb{R}.$$

- For $x < y$, the above condition is equivalent to

$$F_X(x) = F_X(x) F_X(y) \iff F_X(x) = 0 \text{ or } F_X(y) = 1 \text{ or both.}$$

- Similarly, if $y < x$, then we have

$$F_X(y) = F_X(x) F_X(y) \iff F_X(x) = 1 \text{ or } F_X(y) = 0 \text{ or both.}$$

- If $x = y$, then we have the condition

$$F_X(x) = \left(F_X(x)\right)^2 \iff F_X(x) \in \{0, 1\}.$$

Because this property must hold for all $x \in \mathbb{R}$, it follows that X must be a **constant random variable**

Conditional CDF

Definition (Conditional CDF)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X, Y be random variables with respect to $\mathcal{F}$.

1. Fix $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$.

The **conditional CDF of X , conditioned on event A** , is defined as

$$F_{X|A} : \mathbb{R} \rightarrow [0, 1], \quad F_{X|A}(x) := \frac{\mathbb{P}(\{X \leq x\} \cap A)}{\mathbb{P}(A)}, \quad x \in \mathbb{R}.$$

2. The **conditional CDF of X , conditioned on the random variable Y** , is defined as

$$\forall x, y \in \mathbb{R}, \quad F_{X|Y}(x|y) := \frac{F_{X,Y}(x, y)}{F_Y(y)} = \frac{\mathbb{P}(\{X \leq x\} \cap \{Y \leq y\})}{\mathbb{P}(\{Y \leq y\})},$$

whenever denominator is non-zero.

Remark:

- If X, Y are independent, then $F_{X|Y}(x|y) = F_X(x)$ for all x, y such that $F_{X|Y}(x|y)$ is well-defined

Jointly Discrete and Jointly Continuous Random Variables

Jointly Discrete Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that all random variables appearing below are defined with respect to $\mathcal{F}$.

Definition (Jointly Discrete Random Variables)

Random variables $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$ are said to be **jointly discrete** if there exists a discrete set $\mathcal{E} \subset \mathbb{R}^2$ (finite or countably infinite) such that

$$\mathbb{P}\left(\{(X, Y) \in \mathcal{E}\}\right) = \mathbb{P}\left(\{\omega \in \Omega : (X(\omega), Y(\omega)) \in \mathcal{E}\}\right) = 1.$$

Remarks:

- If X and Y are jointly discrete, then X and Y are individually discrete, and vice-versa

Joint PMF

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that all random variables appearing below are defined with respect to $\mathcal{F}$.

Definition (Joint PMF)

The **joint PMF** of random variables X, Y is a function $p_{X,Y} : \mathbb{R}^2 \rightarrow [0, 1]$ defined as

$$p_{X,Y}(x, y) = \mathbb{P}(\{X = x\} \cap \{Y = y\}), \quad x, y \in \mathbb{R}.$$

Remark:

- The joint PMF can be defined for any two random variables X, Y (not necessarily jointly discrete)
- If X, Y are jointly discrete with $\mathbb{P}(\{(X, Y) \in \mathcal{E}\}) = 1$, then

$$\sum_{(x,y) \in \mathcal{E}} p_{X,Y}(x, y) = 1.$$

Properties of Joint PMF

Theorem 1 (Properties of Joint PMF of Jointly Discrete Random Variables)

Let X and Y be jointly discrete random variables with

$$\mathbb{P}\left(\{(X, Y) \in \mathcal{E}\}\right) = \sum_{(x, y) \in \mathcal{E}} p_{X, Y}(x, y) = 1.$$

Furthermore, let $\mathcal{X}$ and $\mathcal{Y}$ be the projections of $\mathcal{E}$ onto the x -axis and y -axis respectively, i.e.,

$$\mathcal{X} = \{x \in \mathbb{R} : (x, y) \in \mathcal{E}\}, \quad \mathcal{Y} = \{y \in \mathbb{R} : (x, y) \in \mathcal{E}\}.$$

Then, the joint PMF $p_{X, Y}$ satisfies the following properties.

1. $\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X, Y}(x, y) = 1.$
2. $\sum_{y \in \mathcal{Y}} p_{X, Y}(x, y) = p_X(x)$ for all $x \in \mathbb{R}.$
3. $\sum_{x \in \mathcal{X}} p_{X, Y}(x, y) = p_Y(y)$ for all $y \in \mathbb{R}.$

Conditional PMF

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that all random variables appearing below are defined with respect to $\mathcal{F}$.

Definition (Conditional PMF)

Let X and Y be **jointly discrete** random variables.

1. Fix $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$.

The **conditional PMF of X , conditioned on event A** , is defined as

$$p_{X|A} : \mathbb{R} \rightarrow [0, 1], \quad p_{X|A}(x) := \frac{\mathbb{P}(\{X = x\} \cap A)}{\mathbb{P}(A)}, \quad x \in \mathbb{R}.$$

2. The **conditional PMF of X , conditioned on the random variable Y** , is defined as

$$\forall x, y \in \mathbb{R}, \quad p_{X|\{Y=y\}}(x) := \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{\mathbb{P}(\{X = x\} \cap \{Y = y\})}{\mathbb{P}(\{Y = y\})},$$

whenever denominator is non-zero.